
AI-Powered Collections Strategy

Tata GenAI Powered Data Analytics Virtual Experience

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Program

B.Tech CSE | 3CSE7

AI-Powered Collections Workflow

DATA TO DECISION



1. Dynamic Inputs

Real-time financial data ingestion.
Credit utilization monitoring.
Debt-to-Income (DTI) ratio tracking.



2. ML Decision Logic

Tree-based ensemble models.
Predict delinquency probability.
Segment into Low, Medium, High risk.

Agentic AI vs. Human-in-the-Loop

SEGMENTED INTERVENTION STRATEGY

Autonomous AI

Dispatching automated digital reminders.

Generating dynamic settlement offers.

Continuous model retraining on outcomes.

Handling routine chatbot inquiries.

Human Oversight

Managing high-risk escalations immediately.

Resolving complex customer disputes.

Approving severe account restructuring.

Auditing AI models for persistent bias.

Responsible AI Guardrails

ETHICAL & REGULATORY SAFEGUARDS

Fairness

Routine statistical auditing to achieve demographic parity and prevent systemic discrimination against protected classes.

Explainability

Implementation of SHAP values to provide transparent, interpretable reasons for individual algorithmic risk scores.

Compliance

Strict programmed adherence to fair debt collection practices and mandatory human override options for customers.

Expected Business Impact

-20%

Operational cost-to-collect

+15%

Early-stage recovery rates

Brand

Preserved loyalty & reduced anxiety